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Assessment

Classification Task Analysis Report

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Abstract

Purpose: This report aims to analyze the Gender Development Groups on a global basis by examining different factors affecting the development of genders.

Approach: The dataset used for the task is Gender Development Index 2019, which measures gaps in human development achievements based on genders. The dataset aligns with the Sustainable Development Goal (SDG) 5: Gender Equality. The dataset consists of 14 columns and 189 rows, each of them representing a country. The dataset is examined to contain null or duplicate values, performed necessary Exploratory Data Analysis (EDA) and created two different models for predicting the target.

Key Results: Both the models were fairly evaluated on the same dataset using Confusion Matrix, Precision, Recall and F1 score. Both the models performed satisfactory for the regression task. The logistic regression (SoftMax) model had a precision value of around 0.61 and the KNN regression model had precision value of around 0.68.

Conclusion: The final classification model performed excellently with a precision value of 0.79, recall value of 0.31 and f1 score of 0.23. The low recall value indicates higher false negative predictions and low F1 score indicates the imbalance between precision and recall value.

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1. Introduction

1.1. Problem Statement

The goal of this project is to develop a machine learning model to predict the Gender Development Groups using key features such as Life Expectancy, Expected years of schooling and Human Development Index of both the genders, male and female.

1.2. Dataset

The project uses a dataset, Gender Development Index 2019 available in Kaggle. The dataset consists of 189 rows and 14 columns. Each of the rows represents a country. This dataset consists of key features to identify Development Index, each for both the genders. The dataset aligns with the Sustainable Development Goal 5: Gender Equality as it identifies the development achievements of both the genders.

1.3. Objective

The objective of this task is to build two different models for classification tasks that can predict the GDI groups from 1 to 5 from each of the country with the help of other independent variables. After that, both the models need to be evaluated and compared using Confusion matrix, precision, recall and F1 score to find the best model for the dataset.

2. Methodology

2.1. Data Preprocessing and Exploratory Data Analysis

Before building the model, it is required to analyze whether the dataset is usable or not. If not, the dataset needs to be made usable. It is important to perform data preprocessing and Exploratory Data Analysis (EDA). Firstly, the dataset was loaded into a dataframe. To make the data usable, the data was first analyzed by using different EDA techniques. One of the major issues in the dataset is that it had “..” values instead of NAN values and the independent columns were of type object. To fix this, the “..” values were first converted to NAN. It was also seen that the data had commas, for example 8900.40 was written as 8,900.40 so the commas were also removed. After that the independent variables were converted to float type while the dependent variable was converted to int type. The NAN values were then filled with the forward Fill.

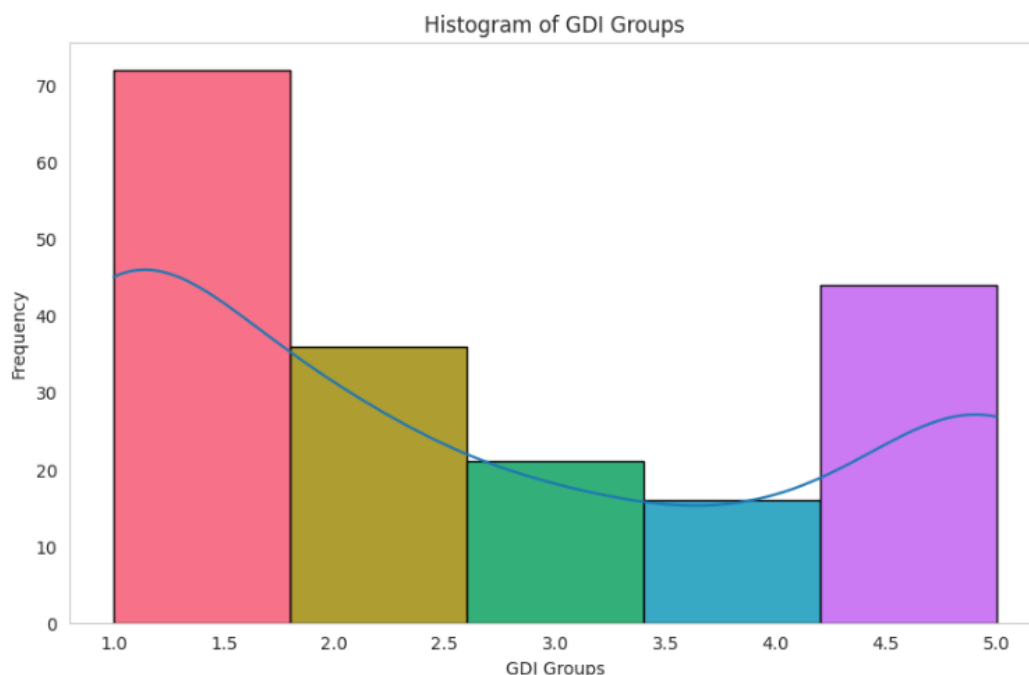


Figure 1Histogram of GDI Groups

The above figure is a histogram representing the GDI Value (Target Variable). This illustrates the distribution of the target variable. The histogram demonstrates that class 1 has the highest frequency among the data while class 4 has the lowest frequency.

After making the data usable, it was prepared for the model learning process. Firstly, independent columns from the data were selected for features and the “GDI_Group” column was selected for target. The data is then split into training and testing sets. The features are then scaled using normalization and one hot encoding was used for the target values.

2.2. Model Building

The scaled features are used to build two different models to predict the one hot encoded target variable. One model was for Logistic Regression using softmax and one for KNN Classification. For this, different functions were created to make the model reusable and scalable in the future.

Logistic Regression Model (Softmax)

Softmax Regression Model depends on the gradient descent to optimize the parameters weights (w) and bias (b). In this model, it is necessary to compute the cost of the model which is the average of the log loss function.

To create the model

- i. Z Prediction: To predict y we need to first find z by using $z = w.x + b$ where, x is independent variable; w is weight and b is bias, and find its exponential and normalize it so that the predicted values are in the range of 0 to 1 and the sum of all predicted values is equal to 1.
- ii. Cost Computation: For the cost, a log loss is calculated and then its average is taken as the cost.
- iii. Gradient Descent: The gradient of the cost is performed and using a learning rate (α), the parameters w and b are updated. After a fixed number of iterations, the gradient descent stops, and the optimal value of w and b is obtained where the cost is minimized.
- iv. Using the optimal value: The optimal values of w and b were used to predict the test set.

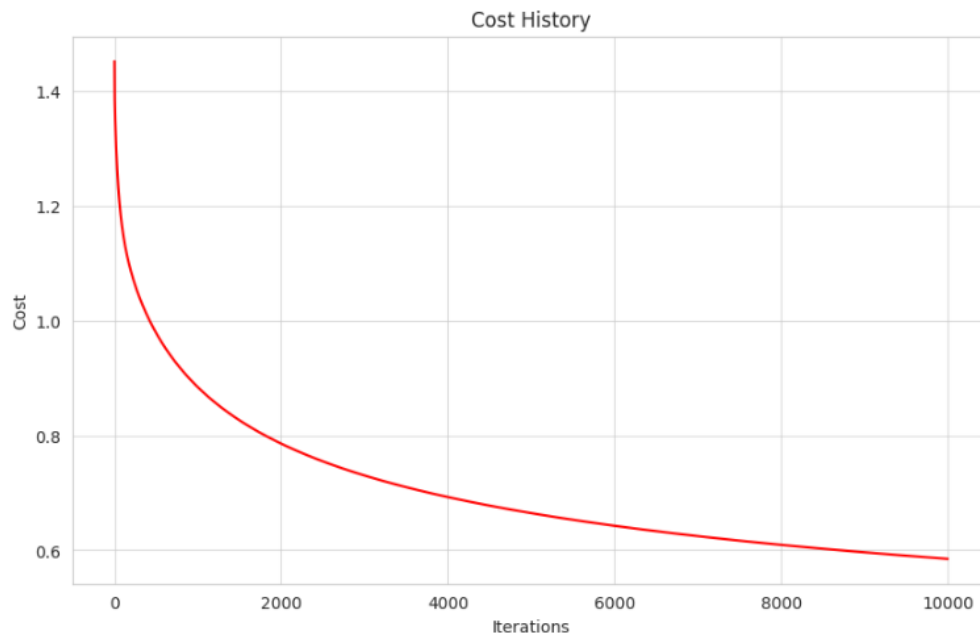


Figure 2 Cost History of Softmax Regression Before Feature Selection

The above graph shows the difference in cost as the number of iterations increases. It can be observed that the cost decreases as the number of iterations increases. The cost descends slower after a certain number of iterations are reached.

KNN Classification Model

K Nearest Neighbors (KNN) Classification Model depends on the k number of its neighbors to predict the value. The model finds the nearest neighbors and finds the mode of those neighbors. The mode is taken as the predicted class.

To create the model:

- Computing Distance: The distance between the data points and the predicting class is computed using Euclidean Distance formula.
- Predict KNN: Using the given value of k, the mode of the k nearest neighbors is found and it is used as prediction class.

2.3. Model Evaluation

The models were evaluated using Confusion Matrix, precision score, recall score and F1 score.

Logistic Regression Model (Softmax)

Precision score: 0.614

Recall score: 0.547

F1 score: 0.495

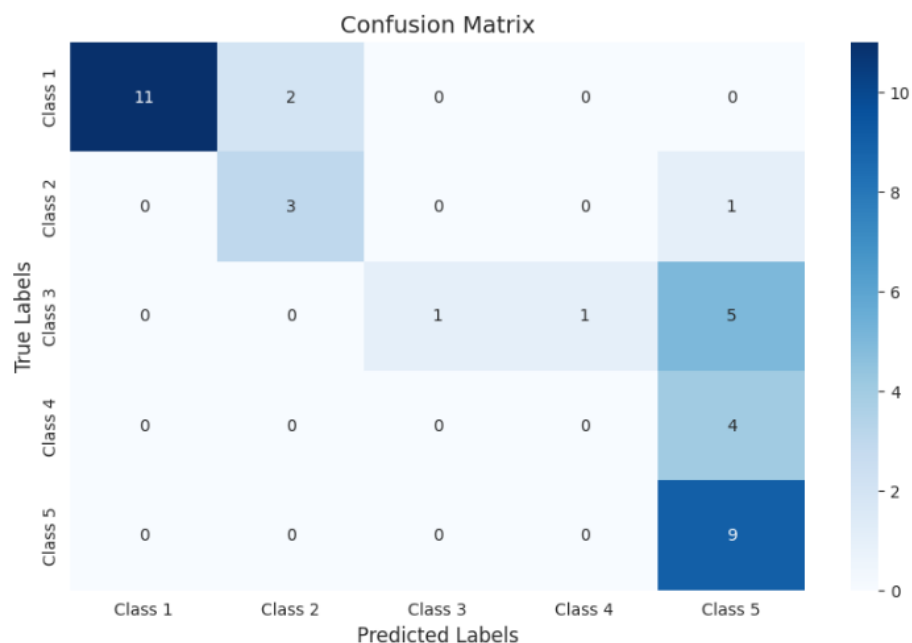


Figure 3 Confusion Matrix of Softmax Regression without Feature Selection

The slightly above average precision score represents higher true positives whereas the recall score represents balance in true negatives and false negatives. The average F1 score represents the balance between precision score and recall score. The model was evaluated on $\alpha = 0.5$ and iteration = 10000.

KNN Classification Model

Precision score: 0.681

Recall score: 0.597

F1 score: 0.5601

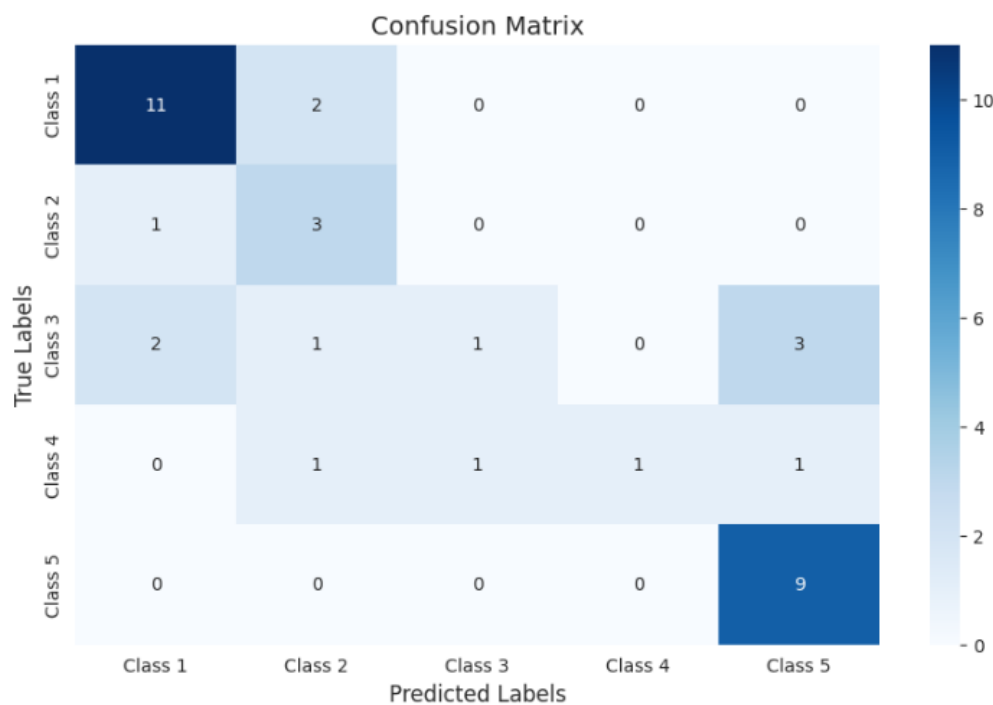


Figure 4 Confusion Matrix of KNN Classification without Feature Selection

The high precision score represents higher true positives and above average recall score represents slightly high true negatives. The F1 score being 0.56 represents an acceptable balance between precision score and recall score. The model was evaluated on $k = 3$.

2.4. Hyper-parameter Optimization

For hyperparameter optimization, a grid search algorithm was implemented. To find the best hyperparameter, it was necessary to use grid search instead of random search despite its performance benefits as the model was highly inconsistent with the use of random search.

Logistic Regression Model (Softmax):

For logistic regression, the hyperparameters to determine are alpha (learning rate) and iteration (iteration count for gradient descent).

For hyperparameter tuning, the following lists were used.

`alpha_list = [0.01, 0.1, 0.2, 0.3, 0.4, 0.5, 1]`

`iterations_list = [1000, 5000, 10000, 20000]`

After using grid search, the best hyperparameters were:

- i. Alpha = 0.01
- ii. Iterations = 5000

The model had a precision of 0.71.

KNN Classification Model:

For KNN classification, the hyperparameter to determine is k (number of nearest neighbors to select).

For hyperparameter tuning, the following list was used.

`k_list = [3, 5, 7, 9, 11, 13, 15, 17]`

After using grid search, the best hyperparameter was:

- i. K = 3

The model had a precision of 0.68.

2.5. Feature Selection

SelectKBest with Chi-Squared score function provided by scikit learn was used for feature selection. The total number of usable features was 11 for classification tasks out of which 7 were selected.

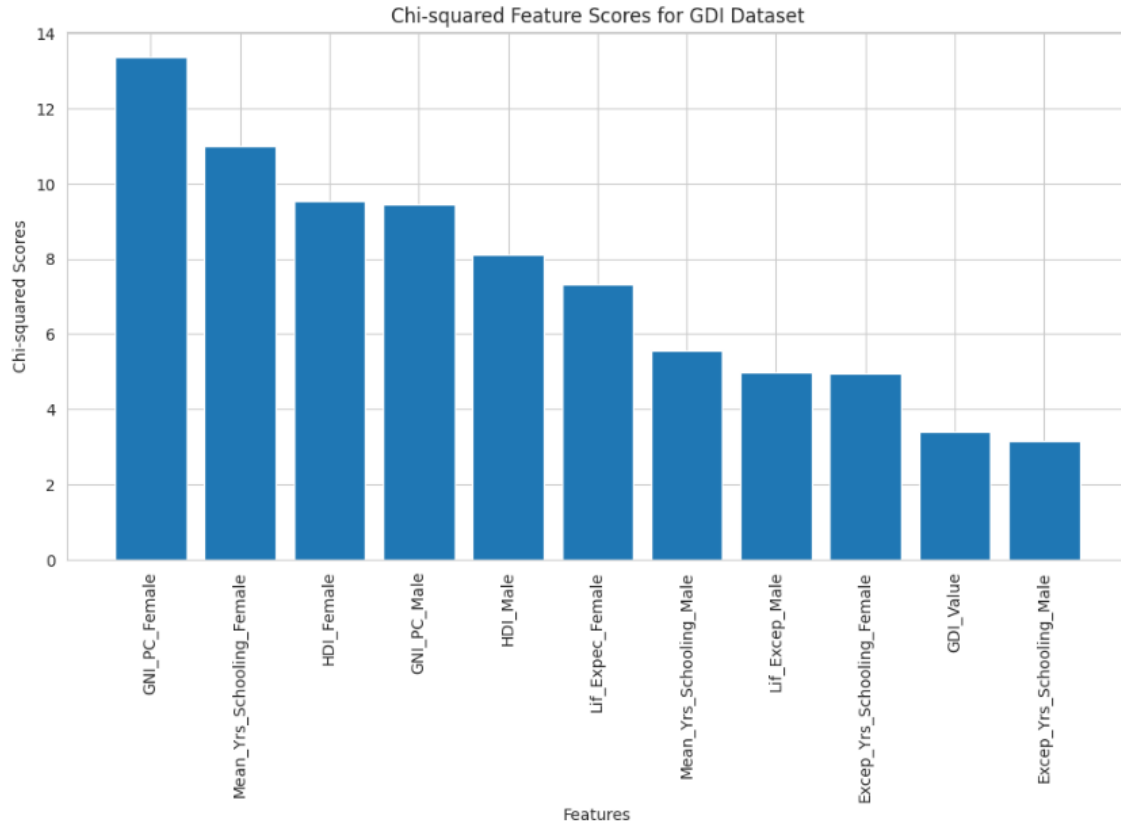


Figure 5 Correlation Scores of Features

The above figure describes the correlation of all 11 features with the GDI groups. The GNI_PC_Female has the highest relation with the GDI value while Excep_Yrs_Schooling_Male value has the lowest relation. Even though the relation is low, it is still usable as an independent variable for this classification task.

Logistic Regression After Feature Selection with Hyperparameter Tuning:

Precision Score: 0.79

Recall Score: 0.31

F1 Score: 0.23

KNN Classification After Feature Selection with Hyperparameter Tuning:

Precision Score: 0.46

Recall Score: 0.33

F1 Score: 0.29

3. Conclusion

3.1. Key Findings

After evaluating the models with hyperparameter tuning and feature selection, it was seen that the performance of the model depends on its hyperparameters and its features. There was a high increase in the precision of the Softmax Regression model after finding the best hyperparameters while there was no noticeable change for KNN Classification task

3.2. Final Model

For this dataset, Logistic Regression Model (Softmax) was used as its final model. After comparing the evaluation of the precision score of both the models after feature selection and hyperparameter tuning in test set, Softmax Regression had a better score with 0.79 while KNN had a score of 0.46.

Building Logistic Regression as Final Model:

To build the Logistic Regression model, the best hyperparameter was used with feature selection.

Hyperparameters:

- i. Alpha = 0.01
- ii. Iterations = 5000

Features Selected: 7

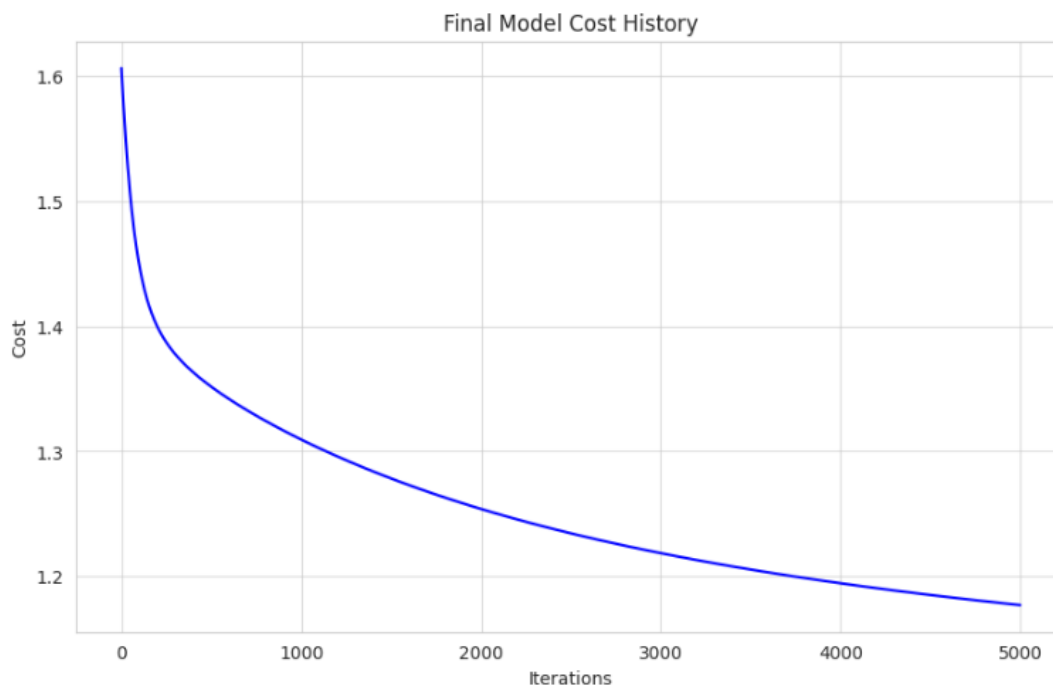


Figure 6 Cost History of Final Model

The figure shows the behavior of the cost function of the final model. The cost decreases as iteration increases. The final cost was 1.176975 from initial cost of 1.606669.

Precision score: 0.79

Recall score: 0.31

F1 score: 0.23

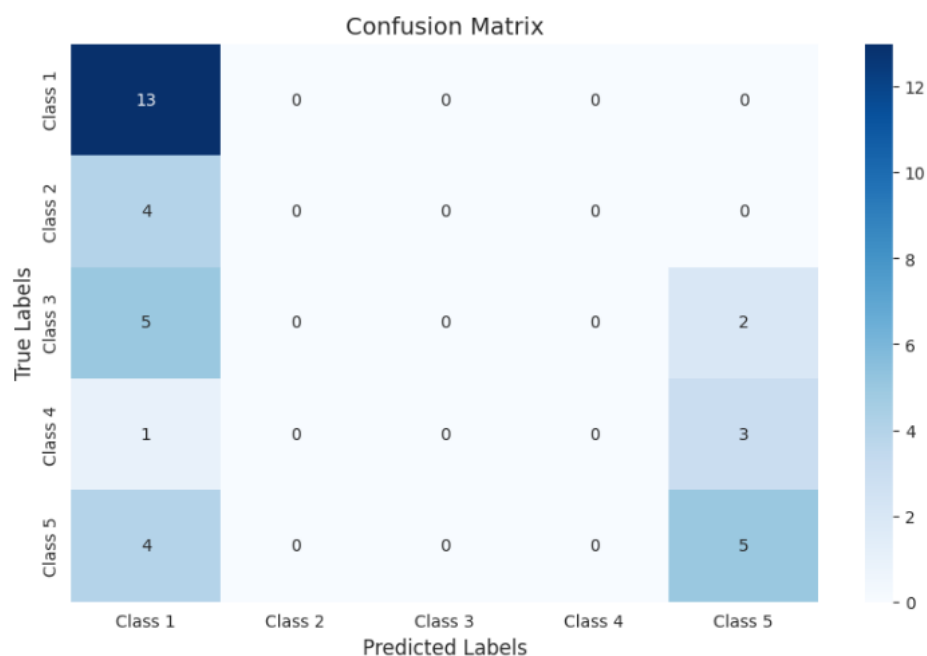


Figure 7 Confusion Matrix of Final Model

The high precision score of 0.79 indicates a high rate of true positives but the low recall score of 0.31 indicates a high rate of false negative leading to low F1 score.

3.3. Challenges

One of the challenges faced was during data analysis. Since the data had “..” instead of NAN values, it was providing a problem for any manual mathematical computations. The dataset required to convert “..” into NAN values and then fill the NAN values using forward fill. Another major challenge faced was the predictions being inconsistent as the hyperparameters changed. To fix this problem, it was necessary to use grid search as random search was having the predictions very inconsistent.

3.4. Future Work

The model has been able to predict the values more precisely and accurately than expected. But in future, instead of Grid Search which consumes a lot of computational cost, more optimized hyperparameter tuning algorithms can be explored to obtain optimal hyperparameters for building a better model.

4. Discussion

4.1. Model Performance

The performance of the models depended on Precision score, Recall score and F1 score. Comparing both models, Logistic Regression Model performed better than KNN Model with precision of 0.79, recall of 0.31 and F1 of 0.23. Overall, both models performed well for this dataset, but Logistic Regression model worked better so it was chosen to build a final model using the selected features and hyperparameters.

4.2. Impact of Hyperparameter Tuning and Feature Selection

After building the final model using the obtained set of hyperparameters and features, the performance of the model improved as precision, recall and F1 score increased. This means that the model is performing well and fitting the data properly.

4.3. Interpretation of Results

The final interpretation for the output obtained from all the models on test set are:

For Logistic Regression Model:

Precision score: 0.614

Recall score: 0.547

F1 score: 0.495

For KNN Classification Model:

Precision score: 0.681

Recall score: 0.597

F1 score: 0.5601

4.4. Limitations

Due to less data being presented in the dataset, the model may be overfitted due to less data to perform training and testing tasks. The total data present in the data set was 189 and out of them 22 rows of data consisted of missing values. Forward fill was used to fill those missing values which may lead to inconsistency and less accuracy in data evaluation.

4.5. Suggestions

Better algorithms for hyperparameter tuning can be conducted and analyzed to reduce computational cost. More data can be collected to train the model to reduce overfitting and fill in the missing values for better accuracy in predicting values.